Go 3 · Research

What is still unsolved, and how to solve it

The research landscape from max-pressure to language-model traffic agents, the eleven gaps that keep them off real streets, five small SUMO experiments that make those gaps visible, and the designs and questions that look most promising.

smrahmanbd.github.io/agentic-atsc-lab

6 ideas, each building on the last

1 The research landscape: six waves

0 of 9 LLM controller papers reporting a field deployment

2 How to evaluate a new idea fairly

0.66 s R1 talking benefit, no bigger than the 0.67 s noise

3 The eleven gaps

53 → 22 s R4 longest wait with fairness weight 0.15

4 Our five experiments, R1-R5

223 vs 0 unsafe commands reaching the signal, R2 at 40 % bad answers

5 Future solution: a layered agentic controller

1.81 M decisions per hour for 2,510 lights deciding every 5 s

6 Open research questions

+26.3 % R5 delay at H = 8 s, for 8.7 % fewer stops

Three surprises to start

Not one LLM traffic controller is on a real street.

The nine language-model controllers in our reviewed literature were all tested in simulation only; none reports a field deployment. Their headline gains, about 17.5-20.4 % in waiting or travel time, are as reported by their authors.

Breaking the rules looks faster.

In our experiment R2, a perfectly reliable model with no safety shield had less delay (9.0 s) than the shielded one (10.6 s). It got there by issuing 112 rule-breaking commands. A fair comparison makes everyone obey the same rules.

A promise can cost more than it saves.

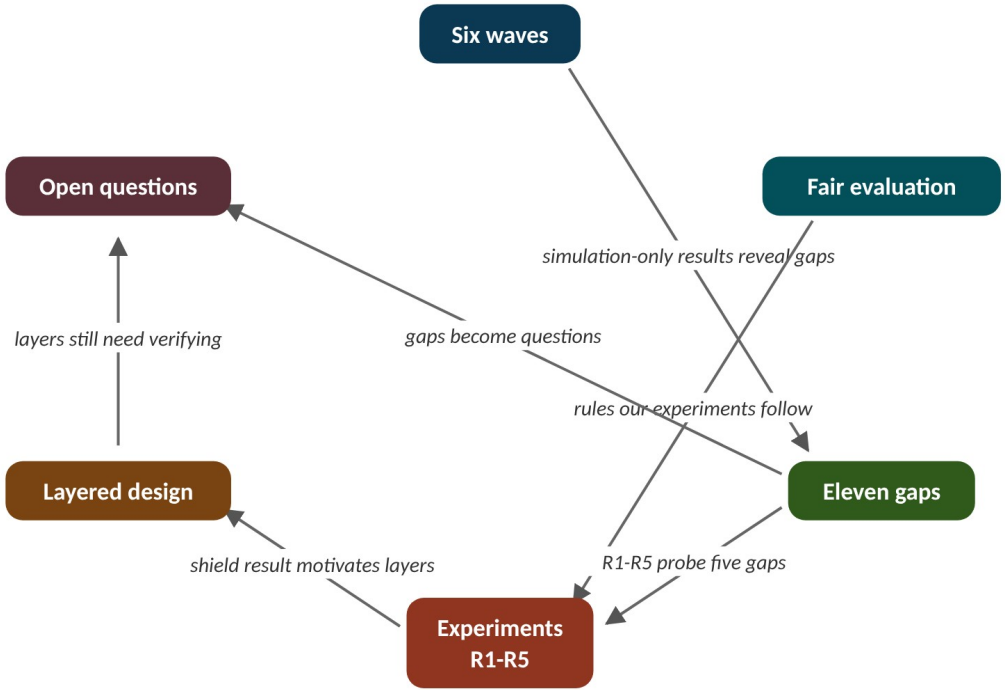
In R5, making the signal commit to its next change 8 s ahead gave connected cars usable speed advice and the fewest stops, but raised delay from 16.7 to 21.1 s and CO2 from 229.5 to 241.4 g per vehicle.

Words you will meet

Reinforcement learning (RL)	A program that learns by trial and error, rewarded with a score, like a dog trained with treats.
Large language model (LLM)	A text-predicting AI, as in chatbots. It can answer 'switch to north-south' and give a reason.
Agent	A program that observes, decides and acts on its own. Here, a signal controller.
Hallucination	An LLM answer that is confident but wrong, such as a phase that does not exist.
Safety shield	A strict rule-checker that blocks any command breaking minimum green or other safety rules.

Sim-to-real gap	Everything that changes when a controller moves from a simulator (SUMO, CityFlow) to real streets.
Seed	The starting number for simulated random arrivals: a different seed is a different, equally likely day.
CAV / V2X	Connected and automated vehicle; V2X is the radio link between vehicles and the signal.
GLOSA	Green Light Optimal Speed Advisory: the speed that arrives on green, so the driver need not stop.

The map of today's ideas



1 The research landscape: six waves

IN PLAIN WORDS

Think of how phones grew up: landline, mobile, smartphone, voice assistant. Each new wave kept the old one inside it: a voice assistant still makes plain phone calls. Newer signal controllers likewise use older controllers as tools or as the yardstick.

- Adaptive, max-pressure, deep RL
- LLM: tool user, controller, hybrid
- Each wave keeps the last as baseline
- Reported LLM gains: about 17.5-20.4 %
- All LLM controllers: simulation only

0 of 9

LLM controller papers reporting a field deployment

Worked example: What does '20.4 % less waiting' mean?

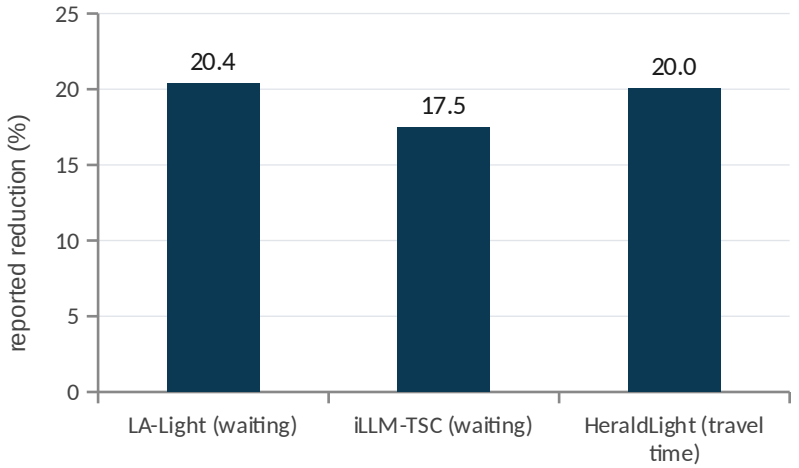
$$\text{new value} = \text{old value} \times (1 - \text{gain} \div 100)$$
$$\text{relative change} = (\text{new} - \text{old}) \div \text{old} \times 100\ \%$$

LA-Light reports 20.4 % less waiting than RL under sensor outages. Suppose (invented) RL waited 50 s.

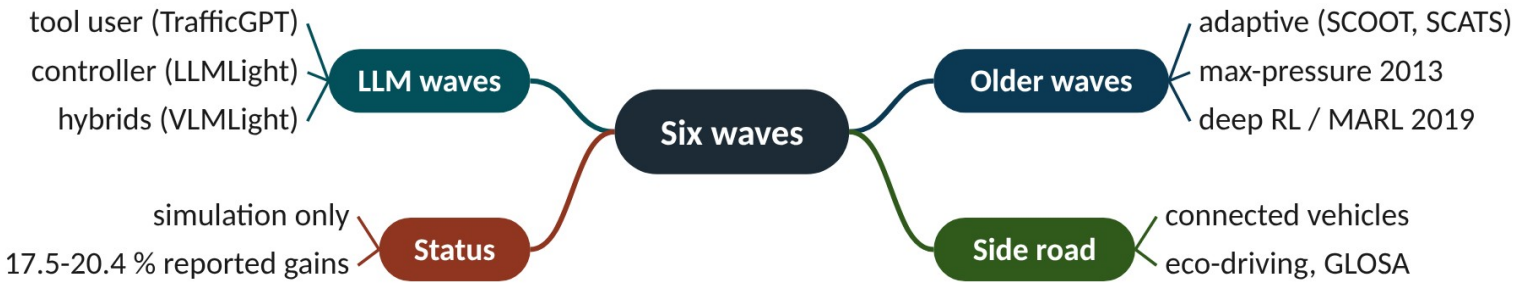
1. Turn the percentage into a fraction: 20.4 ÷ 100 = **0.204**.
2. Seconds saved: 50 × 0.204 = **10.2 s**.
3. New waiting time: 50 − 10.2 = **39.8 s** (or 50 × 0.796).
4. Check backwards: (39.8 − 50) ÷ 50 × 100 = **−20.4 %**.
5. iLLM-TSC reports 17.5 % under degraded communication: 50 × 0.825 = **41.25 s**.

The two gains differ by 1.45 s here, and their setups differ: do not rank them.

Headline gains reported by three LLM papers (authors' simulations).



Remember it: Six waves



PICTURE THIS · STOP 1: MUSEUM OF WAVES

Six bone-dry surfboards: an old wooden SCOOT board, a pressure-gauge board, a neural-network board, a talking board, a board with a brain, a board split in two.

TRAP

A reported percentage holds only for the authors' simulator, data and baseline. Two papers' percentages compare only if the setups match, which is rare.

2 How to evaluate a new idea fairly

IN PLAIN WORDS

A fair race gives every runner the same shoes and track, runs more than one heat, and also checks who was left behind. One heat on a windy day proves little: the winner may just have had the wind at their back.

- Beat strong baselines, not just fixed time
- Same rules for every controller
- Three or more seeds: mean and spread
- Report fairness, CO2 and violations
- Publish the negative results too

0.66 s

R1 talking benefit, no bigger than the 0.67 s noise

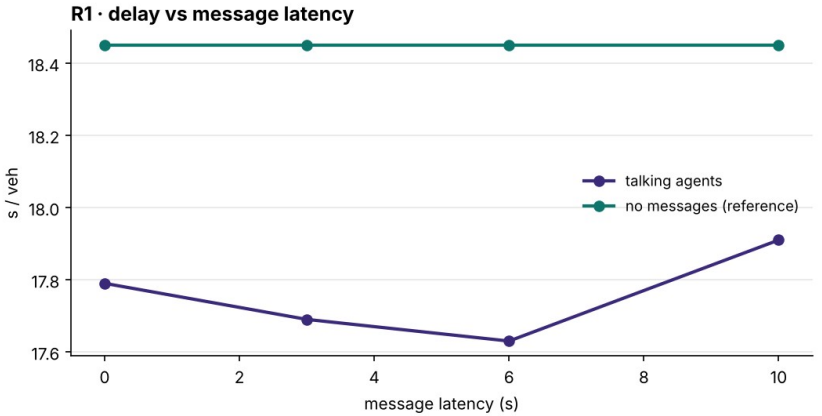
Worked example: R1: is the gain bigger than the wobble?

mean = sum of results ÷ number of runs
range = largest − smallest

R1, four-junction corridor, one seed. No messages: 18.45 s. Six runs with messages (latency 0, 3, 6, 10 s; loss 30 %, 60 %): 17.79, 17.69, 17.63, 17.91, 18.30, 17.86 s. Latency and loss can only hurt, so up-and-down changes behave like noise.

1. Sum: 17.79 + 17.69 + 17.63 + 17.91 + 18.30 + 17.86 = **107.18 s**.
2. Mean: 107.18 ÷ 6 = **17.86 s**.
3. Range: 18.30 − 17.63 = **0.67 s** (standard deviation about 0.24 s).
4. Benefit of talking at 0 s latency: 18.45 − 17.79 = **0.66 s**, that is 3.6 %.
5. Compare: the benefit (0.66 s) is no bigger than the wobble (0.67 s).

With one seed we cannot claim that messages help, or that latency hurts. The honest next step is more seeds, longer corridors and heavier demand.



Remember it: Fair evaluation



PICTURE THIS · STOP 2: RACE TRACK

Referees make every runner use the same shoes, run three races, and time not just speed but also who was left behind.

TRAP

Do not zoom into a chart and read every wiggle as an effect. Check the axis: R1's whole chart covers less than 1 s. A difference smaller than the run-to-run spread is not a result, however neat the line looks.

3 The eleven gaps

IN PLAIN WORDS

A new car that wins on a test track still needs crash tests, rain tests, cheap fuel, a driver who understands the dashboard and a road-safety licence. Speed on the track is only the first step.

- Sim-to-real, safety, hallucination, explainability
- Faults, scale, equity, mixed autonomy
- Predictability, benchmarks, human factors
- R4: fairness weight 0.15 nearly free

53 → 22 s

R4 longest wait with fairness weight 0.15

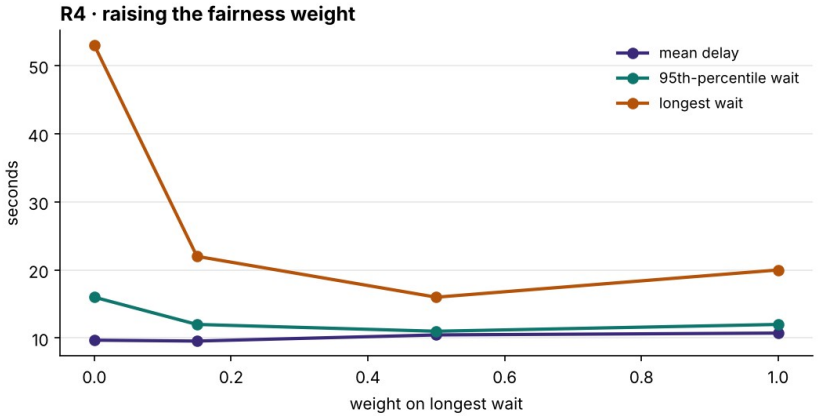
Worked example: R4: what does fairness cost?

fairness-weighted score = efficiency term + w × longest wait
relative change = (new − old) ÷ old × 100 %

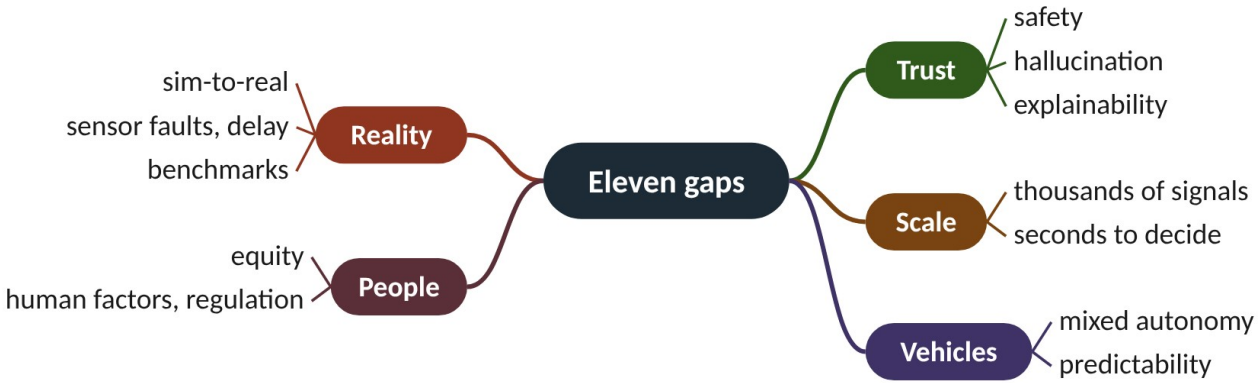
One junction, heavy main road, one seed. Weight w = 0: mean delay 9.69 s, longest wait 53 s. w = 0.15: 9.56 s, 22 s. w = 0.5: 10.46 s, 16 s.

1. Longest wait, w 0 → 0.15: (22 − 53) ÷ 53 × 100 = **−58.5 %**.
2. Mean delay, w 0 → 0.15: (9.56 − 9.69) ÷ 9.69 × 100 = **−1.3 %** (no cost at all).
3. Longest wait, w 0.15 → 0.5: (16 − 22) ÷ 22 × 100 = **−27.3 %**.
4. Mean delay, w 0.15 → 0.5: (10.46 − 9.56) ÷ 9.56 × 100 = **+9.4 %**.
5. Max-pressure (ignores waits): longest wait 150 s, about 150 ÷ 22 = **6.8 times** the w = 0.15 agent.

A small fairness weight was free in this run; larger weights buy a little more fairness for extra delay. One seed only, so treat the 'free' part as a hint, not a proof.



Remember it: Eleven gaps



PICTURE THIS · STOP 3: POTHOLE ROAD

Eleven potholes, each with a sign: a simulator screen, a broken guardrail, a parrot, a question mark, a cut cable, a crowd, a seesaw, blue cars, a broken promise, a single dataset, a judge's gavel.

TRAP

A low mean delay can hide one driver waiting a very long time. Max-pressure had more mean delay here (15.52 s), but the bigger warning is its 150 s longest wait. Always report the tail (95th percentile and maximum), not just the average.

4 Our five experiments, R1-R5

IN PLAIN WORDS

A seat belt slightly annoys you on every trip and does nothing on most of them. On the day of a crash it is the only thing that matters. The safety shield is the seat belt between an LLM and the lights.

- R1: messages help under one second
- R2: shield holds delay at 12.1 s
- R2: no shield, 223 unsafe commands
- R3: CAV-only weaker at 10 %
- R4 and R5: fairness, commitment

223 vs 0

unsafe commands reaching the signal, R2 at 40 % bad answers

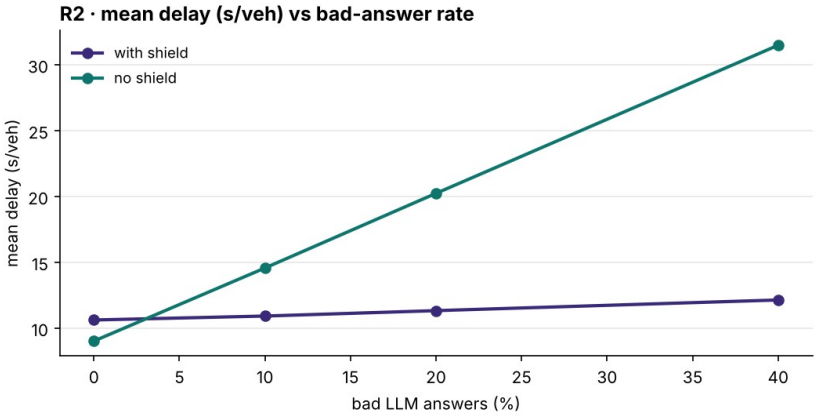
Worked example: R2: the shield at 40 % bad answers

relative change = (with − without) ÷ without × 100 %
times worse = unshielded ÷ shielded

One junction, mock LLM, 40 % bad answers, one seed. With shield: delay 12.14 s, longest wait 39 s. Without: delay 31.49 s, longest wait 132 s.

1. Delay saved: 31.49 − 12.14 = **19.35 s** per vehicle.
2. As a share of the unshielded delay: 19.35 ÷ 31.49 × 100 = **61.4 %** less.
3. Ratio: 31.49 ÷ 12.14 = **2.59**: without the shield, delay is about 2.6 times higher.
4. Longest wait: 132 − 39 = **93 s** shorter with the shield.
5. Shielded agent, 0 % → 40 % bad: (12.14 − 10.62) ÷ 10.62 × 100 = **+14.3 %** only.

With the shield, a model that is wrong 40 % of the time costs only 14 % more delay and breaks no rules. Without it, the junction degrades badly and 223 unsafe commands get through.



Remember it: Experiments R1-R5



PICTURE THIS · STOP 4: FIVE LAB ROOMS

Room 1: tin-can phones with cut strings. Room 2: the parrot without its guard wrecks the junction. Room 3: half-blind goggles. Room 4: a seesaw between speed and fairness. Room 5: a clock that keeps promises.

TRAP

R2 tests the shield, not any real language model: the 'LLM' is a mock that is wrong with a set probability. Do not quote these numbers as how ChatGPT-like models behave at junctions.

5 Future solution: a layered agentic controller

IN PLAIN WORDS

A hospital does not send every patient to the top surgeon. Nurses handle routine cases fast; hard cases go to a specialist; a second doctor checks risky decisions; strict rules forbid dangerous doses; the hospital board sets the goals.

- Fast base policy for routine cases
- Slow LLM only for hard cases
- Critic checks, shield enforces
- Digital twin tests in shadow mode
- People set goals and read reasons

1.81 M

decisions per hour for 2,510 lights deciding every 5 s

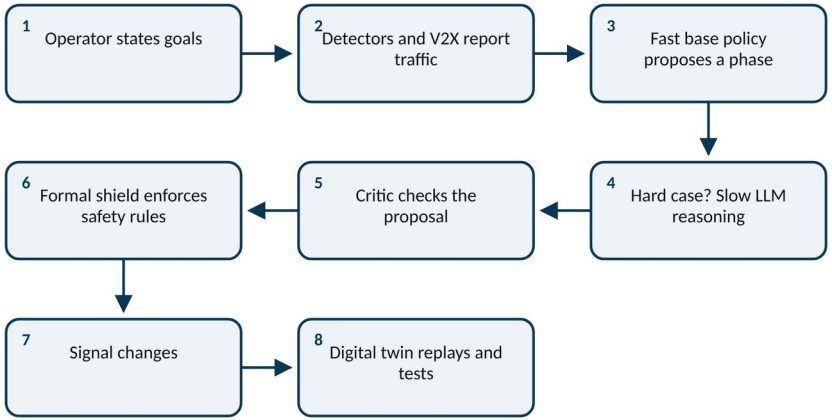
Worked example: Why not ask the LLM every time?

decisions per hour = signals × (3600 ÷ seconds between decisions)
LLM calls per hour = decisions per hour × share sent to the slow layer

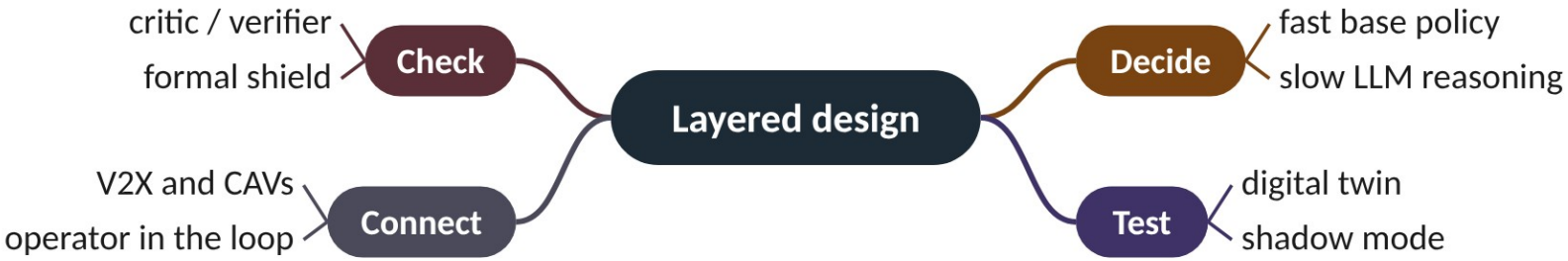
MPLight was demonstrated on 2,510 simulated lights. Suppose (invented settings) each decides every 5 s, and only 1 % of decisions are escalated to the LLM.

1. Decisions per light per hour: 3600 ÷ 5 = **720**.
2. Whole network: 2,510 × 720 = **1,807,200** decisions per hour.
3. Per second: 1,807,200 ÷ 3600 = **502** decisions every second.
4. Escalate only 1 %: 1,807,200 × 0.01 = **18,072** LLM calls per hour.
5. That is about 18,072 ÷ 3600 = **5** LLM calls per second instead of 502.

The fast layer handles 99 % of the work; the slow layer gets 100 times fewer calls. A learned 'when to escalate' rule is an open research question.



Remember it: Layered design



PICTURE THIS · STOP 5: LAYER-CAKE CAFÉ

A seven-layer cake: a fast sponge base, a slow thinking cream, a taster who checks every slice, an iron plate underneath, a mirror cake (digital twin) beside it, radio candles on top, and the chef watching.

TRAP

The shield is not the critic. The critic is a model or checker that can be wrong and gives advice; the shield is fixed, hard-coded rules in the signal cabinet that can never be bypassed. A safe design needs the shield even if the critic is excellent.

6 Open research questions

IN PLAIN WORDS

A lighthouse does not show you the island; it shows where the rocks are so you can sail toward it. A good research question works the same way: it marks clearly what is not known, so a study can head straight for it.

- Can LLM supervision be verified safe?
- Are LLM explanations faithful?
- Which sim-to-real gap dominates?
- Beat actuated below 25 % CAVs?
- R5: fewer stops, more delay, CO2

+26.3 %

R5 delay at H = 8 s, for 8.7 % fewer stops

Worked example: R5: the price of a promise

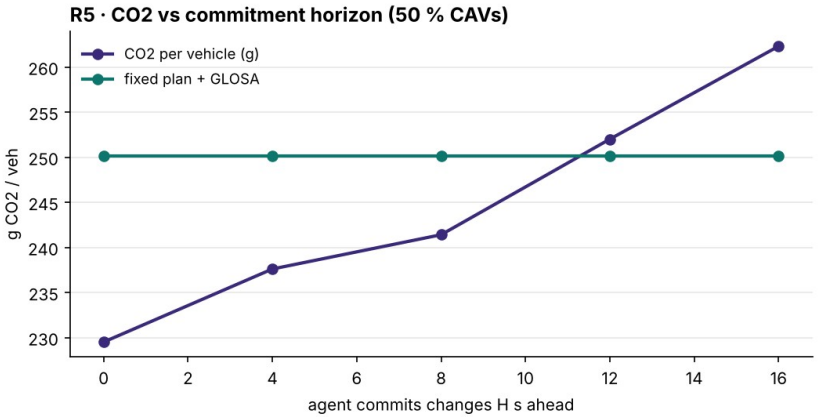
research question = gap + method + setting + metric
relative change = (H value − H 0 value) ÷ H 0 value × 100 %

Corridor, 50 % CAVs with GLOSA, one seed. H = 0 (fully adaptive): delay 16.72 s, CO_2 229.5 g, 0.942 stops. H = 8 s: 21.12 s, 241.4 g, 0.860 stops. H = 16 s: 30.10 s, 262.3 g.

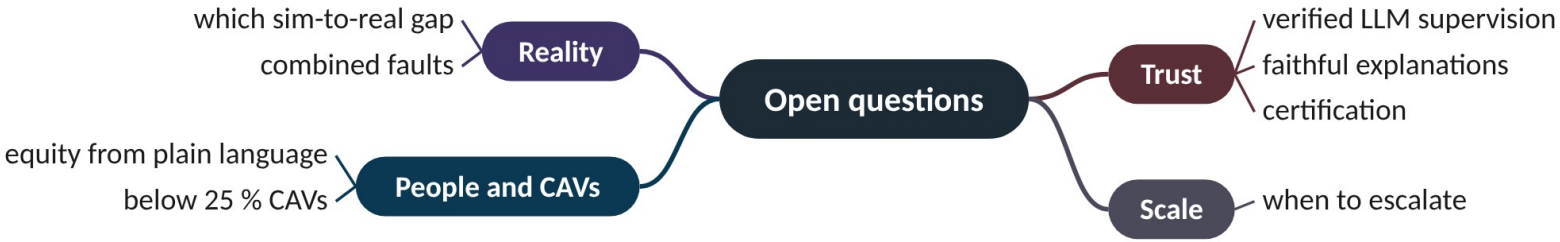
1. Stops, H 0 → 8: (0.860 − 0.942) ÷ 0.942 × 100 = **−8.7 %** (the gain).
2. Delay, H 0 → 8: 21.12 − 16.72 = 4.40 s; 4.40 ÷ 16.72 × 100 = **+26.3 %**.
3. CO_2, H 0 → 8: 241.4 − 229.5 = 11.9 g; 11.9 ÷ 229.5 × 100 = **+5.2 %**.
4. CO_2, H 0 → 16: (262.3 − 229.5) ÷ 229.5 × 100 = **+14.3 %**.
5. Delay, H 0 → 16: (30.10 − 16.72) ÷ 16.72 × 100 = **+80.0 %**.

An 8.7 % cut in stops cost 26 % more delay and 5 % more CO_2.

Longer promises are worse on every count. Predictability without losing adaptivity remains open.



Remember it: Open questions



PICTURE THIS · STOP 6: LIGHTHOUSE

A lighthouse sends eight beams into the fog, each lighting a different unexplored island. On the nearest, a promise-keeping clock stands by a car puffing grey smoke.

TRAP

Fewer stops does not mean less CO2 or less delay. In R5 the 8 s promise gave the fewest agent stops, yet both delay and CO2 went up. Judge eco-driving by every metric, not the one it was designed to improve.

Four families of signal controller

	Max-pressure	Deep RL	LLM controller	Hybrid
Decides with	local queues in minus out	a policy learned from reward	a prompt describing traffic	fast policy + LLM for hard cases
Explains itself?	rule is simple	usually no	yes, a stated reason	yes, for escalated cases
Main worry	starving side streets	sim-to-real transfer	hallucination, latency	when to escalate
Strongest evidence	proof of max throughput; hardware-in-the-loop	simulation, up to 2,510 lights	simulation only	simulation only

Things beginners mix up

This…	…is not this	The fact that tells them apart
RL	**LLM**	RL learns from reward by trial and error; an LLM predicts text and is prompted, not trained per junction.
Critic	**Shield**	A critic is a model giving advice and can be wrong; a shield is fixed rules that are never bypassed.
Seed	**Scenario**	A seed is another random day of the same traffic; a scenario is a different network or demand.
Mean delay	**Longest wait**	R4: mean delay barely moved while the longest wait fell 53 → 22 s.
Fewer stops	**Less CO2**	R5 at H = 8 s: fewest stops, but CO2 rose 229.5 → 241.4 g.
Reported gain	**Field result**	17.5-20.4 % LLM gains are authors' simulation results; none is from a real street.

Check yourself (1 of 2)

LA-Light reports 20.4 % less waiting than RL. If (invented) RL waiting were 60 s, what would LA-Light's be?

A. 39.6 s

B. 47.8 s

C. 48.0 s

D. 72.2 s

Which experiment probes the mixed-autonomy gap?

A. R1

B. R2

C. R3

D. R4

Answers are in the speaker notes and in the study guide's answer key.

Check yourself (2 of 2)

In the layered design, what is the digital twin used for?

A. Replacing detectors

B. Running new policies in shadow mode and measuring sim-to-real gaps

C. Talking to drivers

D. Enforcing minimum green

Three seeds give a new controller 10, 11, 12 s and the old one 11.5, 12, 12.5 s (invented). What should you report?

A. 'The new one is 1 s better', nothing more

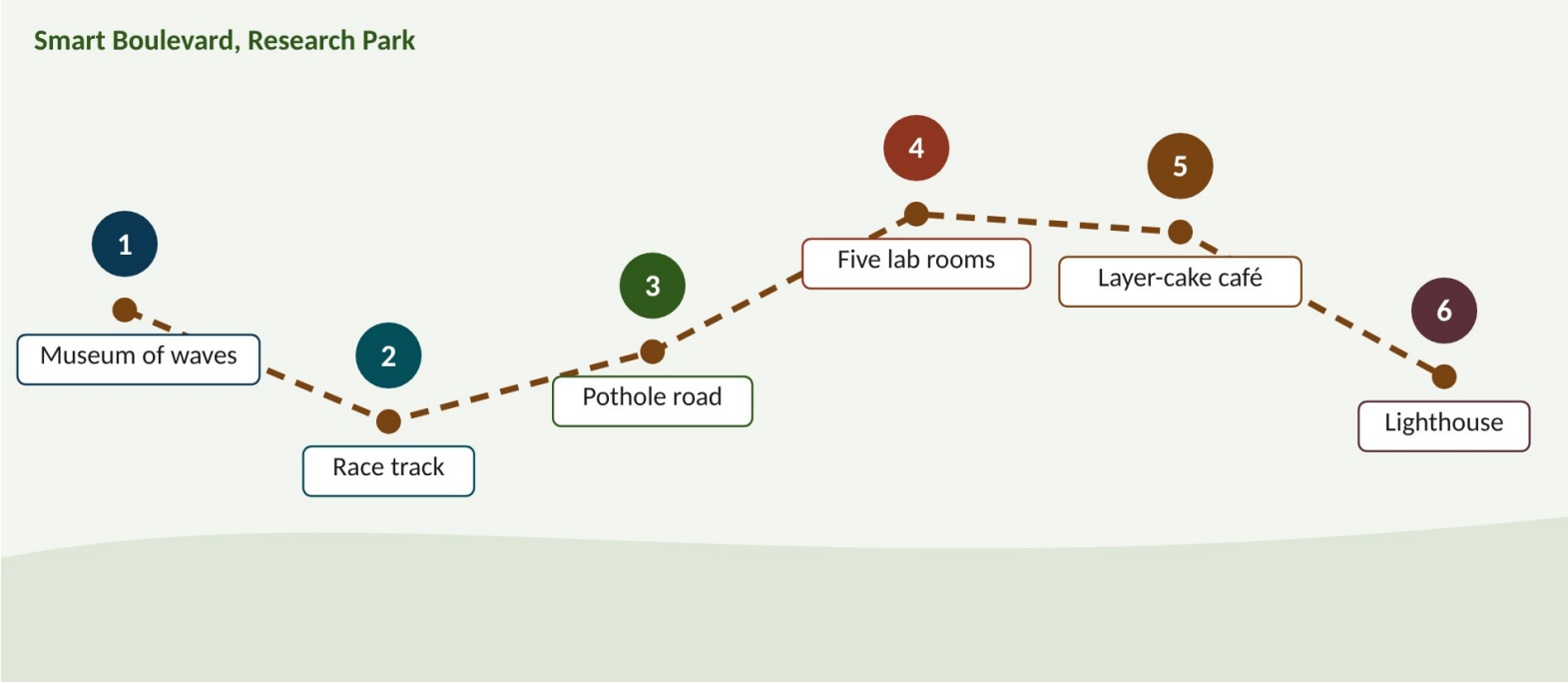
B. Means 11 and 12 s with ranges 2 and 1 s: promising but not shown; run more seeds

C. The old one is better

D. Only the best seed

Answers are in the speaker notes and in the study guide's answer key.

Walk the memory route



1 Museum of waves = Six waves · 2 Race track = Fair evaluation · 3 Pothole road = Eleven gaps · 4 Five lab rooms = Experiments R1-R5 · 5 Layer-cake café = Layered design · 6 Lighthouse = Open questions

Numbers to remember

› LLM gains reported: 17.5 % (iLLM-TSC), 20.4 % (LA-Light), 20.03 % (HeraldLight); 0 of 9 in the field

› MPLight: 2,510 lights; Feng 2015: −16.33 % at high CV penetration

› R1: 18.45 s silent vs 17.79 s talking; runs range 0.67 s (noise)

› R2 at 40 % bad: 12.14 vs 31.49 s; 0 vs 223 violations

› R3 at 10 %: CAV-only 9.94 s vs loops 9.01 s; 100 %: 6.6 s

› R4: w = 0.15 → longest wait 53 → 22 s; max-pressure 150 s

› R5: H = 8 s fewest stops 0.86; delay 16.72 → 21.12 s, CO_2 229.5 → 241.4 g

Keep going

References, experiments and interactive website: smrahmanbd.github.io/agentic-atsc-lab